

Lars Schöbitz - CV

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2026-07-21

Environmental engineer and data scientist working in the global water, sanitation and health (WASH) sector. Open Science Specialist at ETH Zurich. I develop data science training, build open-data communities, and provide consulting to help manage the world's sanitation challenges. My work spans every sector under the sun, which has made me highly adaptable and empathetic.

Contact

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Technical Skills

R / tidyverse

Quarto / R Markdown

Git / GitHub

R Shiny

Reproducible research

SQL

HTML / CSS

Languages

German (native)

English (fluent)

Swedish

French

Work Experience

Open Science Specialist

Zurich, CH

ETH Zurich - Global Health Engineering

2021 - Current

- Support faculty, PhD and MSc students in applying reproducible workflows in their research.
- Oversee reproducibility and open science aspects of published journal papers; manage online and social media channels.

- Set up organisational work, task and project management tools.

Director & Founder

Zurich, CH

Lars Schöbitz GmbH

2018 - Current

- Teaching Data Science with R.
- Consultant for city-wide faecal sludge management systems.
- Supporting organisations in establishing strong monitoring and evaluation frameworks.

Data Science Instructor

Zurich, CH

Statistical Office of the Canton of Zurich

2021

- Teaching a Data Science with R course to civil servants.

Data Science Instructor

Boulder, USA

University of Colorado Boulder

2021

- Developed and taught an online module to teach R.

Data and Impact Expert

Essex, UK

Aquaconsult Limited

2018 - 2021

- Venture Advisor for the Transition to Scale Program in Urban Sanitation by Grand Challenges Canada.
- Support to Clean Team (Ghana), Eau&Vie (Bangladesh) and Safi Sana (Ghana).

Data Science Intern

Zurich, CH

Statistical Office of the Canton of Zurich

2020

- Developed an R Shiny dashboard for analytics of incoming data analysis requests and an R package for web analytics with Matomo.
- Published a fully reproducible news article; supported COVID-19 monitoring and data analysis; organised a session at DINAcon 2020.

Consultant

Duebendorf, CH

Eawag / Sandec

2020

- Supported the preparation of factsheets for a publication on city-wide inclusive sanitation (CWIS) approaches and tools.

Lead Consultant

Cap-Haitien, Haiti

Inter-American Development Bank

2019 - 2020

- Led a team of consultants developing city-wide faecal sludge management systems.
- Detailed review of a design proposal for a faecal sludge treatment plant; preparation of a Shit Flow Diagram (SFD) for Cap-Haitien.

Data Scientist

Putnam, USA

Biomass Controls PBC

2017 - 2020

- Building tools for data visualisation and business intelligence using the R Shiny framework.
- Support of R&D work and preparation of results for publications and conferences.

Data Scientist

London, UK

Gather

2018

- Supported conceptualization of a data dive for urban sanitation data.
- Cleaned and prepared toilet census data (Kanyama compound, Lusaka, Zambia) for analysis.

Consultant

Geneva, CH

World Health Organization

2016 - 2018

- Global monitoring of the sanitation chain: proof of concept testing for monitoring indicators on safely managed sanitation (SDG 6.2) and safely treated wastewater (SDG 6.3).
- Production of country files for SDG 6.3.1 (proportion of wastewater safely treated).

Research Project Manager

Duebendorf, CH

Eawag / Sandec

2012 - 2016

- Project management with global, multi-disciplinary teams (RRR Business Models, SFD Methods and Tools development).
- Research lead on projects developing diagnostic tools for faecal sludge management.
- Mentor, lecturer and networker.

Research Assistant

Durban, ZA

PRG, University of KwaZulu-Natal

2011 - 2012

- Collected water samples from a DEWATS plant; emptied pit latrines; built a rig to separate pit contents into constituents for analysis.

Education

RStudio Instructor Programme

RStudio (Posit)

2019 - 2020

- Certified as an RStudio (Posit) Instructor.

Environmental and Hygiene Engineering (Dipl.-Ing.)

Giessen, DE

Mittelhessen University of Applied Sciences (THM)

2008 - 2012

- Thesis: Performance Evaluation of a full-scale DEWATS plant in South Africa.

ERASMUS Exchange

Umea, SE

Umea University

2010

- Main subject: Northern Aquatic Systems, including sampling of surface water and analysis of DOC, COD, main nutrients and heavy metals.

Change and Project Management

Mikkeli, FI

Mikkeli University of Applied Sciences

2009

Open Data & Software

AGUASAN Workshop 2027 - Publishing WASH Data Together

AGUASAN Community of Practice / SDC Water Network

I build and maintain open-data communities and FAIR data tooling.

2026 - Current

- Author of a topic proposal for a five-day working retreat where WASH-sector participants publish datasets while learning Git/ GitHub, Quarto and safe use of AI agents; bridges academia (ETH), the WASH community and Swiss government.

fairenough - tidy data into documented R packages

Author

2025 - Current

- Converts tidy data into R packages with documentation websites and AI-assisted descriptions.

jmpwashdata - WHO/UNICEF JMP data

Author

2024 - Current

- R package providing tidy access to WHO/UNICEF Joint Monitoring Programme WASH data.

Data Science for openwashdata (ds4owd)

Course Creator & Instructor

2023 - Current

- Designed and taught two free online cohorts on literate programming with Quarto and R; hundreds of participants across many countries.
- Focus on data management, reproducibility and publishing open data.

washr - R package for publishing openwashdata resources

Author

2023 - Current

- An R package that makes publishing openwashdata data resources easier.

openwashdata community

Founder & Community Lead

2022 - Current

- Built and lead a global open-data community for the WASH sector; 90+ open datasets published as documented R data packages.
- Created the {washr} and {fairenough} R packages that turn tidy data into FAIR, documented data packages.
- See openwashdata.org

rstatsZH - R training for the Canton of Zurich

Course Creator & Instructor

2021 - Current

- Built and ran the R training programme for the cantonal administration of Zurich (rstatsZH), including course website and material.

Teaching

I am passionate about teaching reproducible data science to newcomers.

CVEN 5837/5873/5999 - Data Analytics for Development (4 iterations)

Boulder, USA (online)

University of Colorado Boulder

2022 - 2026

- Graduate course teaching R, the tidyverse, Git/GitHub and Quarto for reproducible data analysis in water, air quality and global health; taught online in 2022, 2023, 2025 and 2026.

Data Science for openwashdata (ds4owd), Cohort 2

openwashdata / ETH Zurich

2025-09 - 2025-12

- Free 9-module project-based online course on reproducible WASH data analysis with R and Quarto.

Research Beyond the Lab - Open Science and Research Methods (3 iterations)

Zurich, CH

ETH Zurich

2022 - 2025

- Semester course (151-8102-00L) teaching field-based research methods and open science with the tidyverse, Git, GitHub and Quarto; taught in spring 2022, 2024 and 2025.

Quarto - Authoring and Publishing for Collaborative Scientific Writing

ETH Zurich - Global Health Engineering

2024-04

- Two-day online course on authoring and publishing scholarly documents with Quarto.

Data Science for openwashdata (ds4owd), Cohort 1

openwashdata / ETH Zurich

2023-10 - 2024-02

- Free 10-module online course teaching R, the tidyverse, Git/GitHub and Quarto for reproducible WASH data analysis.

Open Science Tools for Collaborative Scientific Writing

Zurich, CH

ETH Zurich - Global Health Engineering

2023-10

- Two-day in-person course on open science tools (GitHub, ORCID, Zotero, Posit Cloud) for collaborative scientific writing.

rstatsZH - Data Science mit R (8 course iterations)

Zurich, CH

Statistical Office of the Canton of Zurich

2021 - 2023

- Designed and delivered eight in-person iterations (k001-k008) of an introductory data science with R course for staff of the Canton of Zurich administration.

Lecturer - Faecal Sludge Management

Giessen, DE

Mittelhessen University of Applied Sciences (THM)

2017

- Introduction to Faecal Sludge Management for environmental engineering students.

Lecturer - Sanitation MOOC

Massive Open Online Course: Planning & Design of Sanitation Systems and Technologies

2016

- Prepared a module on Shit Flow Diagrams.

Lecturer - Faecal Sludge Management

Switzerland

NADEL (ETH Zurich) & EPF Lausanne

2013 - 2016

- Teaching on the basics of Faecal Sludge Management.

Workshops

Quarto for Collaborative Scientific Writing - RDM Summer School

Zurich, CH

ETH Zurich

2026-06

- Half-day course on Quarto and the RStudio IDE at the ETH Research Data Management Summer School 2026.

Git for Scientists

Zurich, CH

ETH Zurich - Global Health Engineering

2026-07

- Half-day workshop on Git and GitHub as a foundation for AI-assisted, reproducible and reviewable research.

Git for Scientists

Zurich, CH

CIS, ETH Zurich

2025-05

- Workshop on Git and GitHub for scientific collaboration at the CIS PhD retreat.

Quarto for Collaborative Scientific Writing - RDM Summer School

Zurich, CH

ETH Zurich

2024-06

- Half-day course on Quarto and the RStudio IDE at the ETH Research Data Management Summer School 2024.

FAIR Data Publishing

UKZN, South Africa

UKZN WASH Centre

2024-11

- One-day workshop introducing FAIR data publishing and the washr R package.

Data Stewardship - SST Educational Workshop

Basel

Swiss Toxicology Society (SST)

2023-11

- Educational workshop on data stewardship.

Reference Management with Zotero

Hubris to Hope (HS23)

2023-10

- Workshop on reference management with Zotero.

open by default - a data management strategy

Zurich, CH

ETH Research Data Management Summer School

2023-06

- Teaching at the ETH RDM Summer School.

Data Stewardship Network Kick-off

Zurich, CH

ETH Zurich

2023-02

- Connecting people and topics on data stewardship at ETH Zurich.

Talks

Hello, I am a Data Steward - ReproducibiliTea Journal Club

University of Zurich

2026-05

- Talk at the ReproducibiliTea journal club.

Publishing Open Metadata for ORD Projects of the ETH Domain

Swiss Open Academic Data (SOAD) Day

2025-09

- Presentation on the ethord open metadata dataset.

Hello, I am a Data Steward - Career Talk

Research Software Engineering and Data Stewardship Career Panel

2025-06

- Career panel on research software engineering and data stewardship.

Computational Reproducibility Seminar

Swiss Reproducibility Network (SwissRN)

2025-05

- Seminar talk on computational reproducibility.

Plan for tomorrow today - D-USYS Transdisciplinary Lab Brown Bag

ETH Zurich D-USYS TdLab

2025-05

- Brown-bag talk on a model for data stewardship.

Open Research Data Program of the ETH Board - open metadata dataset

ETH ORD Programme

2025-05

- Talk on a proposal for an open metadata dataset for the ETH Board ORD Program.

Plan for tomorrow today - a model for data stewardship

ETH Library 17:15 Kolloquium

2024-11

- Kolloquium talk on a model for data stewardship.

Swiss Research Data Day (SRDD)

SRDD

2024-10

- Contribution to the Swiss Research Data Day.

Data Stewardship at Global Health Engineering

ETH Zurich

2024-09

- Talk on data stewardship at Global Health Engineering.

Plan for tomorrow today - why you need a data steward

Swiss Reproducibility Conference (SwissRN)

2024-06

- Conference presentation on data stewardship.

UNC Water and Health Conference

Chapel Hill, USA

UNC

2016

- Safely-Managed Sanitation: Monitoring higher levels of service in SDG Target 6.2. Event support.

Sandec Research Seminar

Duebendorf, CH

Eawag / Sandec

2016

- Shit-Flow-Diagrams and the potential for monitoring of Sustainable Development Goals.

UNC Water and Health Conference

Chapel Hill, USA

UNC

2015

- City level excreta flow analysis; the SFD Promotion Initiative. Poster.

Stockholm World Water Week

Stockholm, SE

SIWI

2015

- Side events: Liveable Cities for All; SFD as a tool to foster sustainable urban sanitation programming; Sustainable City Sanitation from planning to implementation.

4th AfricaSan Conference

Dakar, SN

AfricaSan

2015

- Side event: SFD a tool to boost urban sanitation (presentation).
Panellist: Public-Private Partnerships to Meet the Sanitation Shortfall.

3rd Faecal Sludge Management Conference

Hanoi, VN

FSM3

2015

- Quantification of FS by truck-counting studies with the Private Emptiers Association in Kampala (presentation).
- Scaling-up sanitation businesses in low- and middle-income countries (presentation); session moderation; workshop facilitation on FSM systems approach.

37th WEDC International Conference

Hanoi, VN

WEDC

2014

- FAQ: Faecal Sludge Quantification and Characterization - field trial of methodology in Hanoi, Vietnam (conference paper).

Stockholm World Water Week

Stockholm, SE

SIWI

2014

- Side event: From Faecal Sludge to Fuel - Safe Sanitation with Business Opportunities.

36th WEDC International Conference

Nakuru, KE

WEDC

2013

- Espresso Shot Presentation on Resource, Recovery & Reuse.

2nd Faecal Sludge Management Conference

Durban, ZA

FSM2

2012

- Business model innovations for scaling-up faecal sludge businesses in developing countries (abstract & presentation).

Water Institute of Southern Africa (WISA) Conference

Cape Town, ZA

WISA

2012

- Practical experience and operating data from a full-scale decentralised wastewater treatment plant treating domestic wastewater.

Publications

Light absorbing carbonaceous aerosols: Concentrations and sources in an under-monitored region

Ecotoxicology and Environmental Safety

2025

- Co-author. Peer-reviewed journal article.

Estimating Safely Managed Sanitation in Urban Areas; Lessons Learned From a Global Implementation of Excreta-Flow Diagrams

Frontiers in Environmental Science

2020

- Peer-reviewed journal article on SDG 6.2 monitoring.

Towards an off-grid fecal sludge treatment unit: demonstrating energy positive thermal treatment

Gates Open Research

2019

- Peer-reviewed journal article.

Methods to reliably estimate faecal sludge quantities and qualities for the design of treatment infrastructure

Journal of Environmental Management

2018

- Peer-reviewed journal article.

GIS Analysis and Optimisation of Faecal Sludge Logistics at City-Wide Scale in Kampala, Uganda

Sustainability

2017

- Peer-reviewed journal article.

FAQ: Faecal sludge quantification and characterization - field trial of methodology in Kampala, Uganda

Sandec News

2014

- Journal article.

Conference Papers

Remote Sensing, Mobile Applications and Open Data Science Tools for Better Monitoring and Evaluation of Sanitation Services

HIC 2018, 13th International Conference on Hydroinformatics

2018

- Conference paper.

FAQ: Faecal Sludge Quantification and Characterization - field trial of methodology

37th WEDC International Conference

2014

- Conference paper.

Technical Reports

SFD Promotion Initiative - Final Reports (7 cities)

Sustainable Sanitation Alliance / Eawag-Sandec

2015 - 2016

- Lead / co-author of Shit Flow Diagram (SFD) reports for Bignona (SN), Hanoi & Danang (VN), Kampala (UG), Dar es Salaam (TZ), Khulna (BD), Nonthaburi (TH).

RRR-Project: From Research to Implementation, Component 1 - Waste Supply and Availability

Eawag-Sandec

2014

- Technical report on resource recovery and reuse.

Professional Roles

Local Node Leader, Swiss Reproducibility Network (SwissRN)

ETH Zurich - Global Health Engineering

2024 - Current

- Local Node Leader for ETH Zurich in SwissRN, promoting reproducible and open research practices across the institution.

Member, Data Stewardship Network (DSN)

ETH Zurich

2023 - Current

- Data steward for the Global Health Engineering group in ETH Zurich's Data Stewardship Network, an ETH Library / swissuniversities ORD initiative embedding Open Research Data practices across research groups.

References

Dr. Christian Zurbrügg

Duebendorf, CH

Eawag / Sandec

N/A

- Member of the Directorate & Group Leader Solid Waste Management.

Prof. C. A. Buckley

Durban, ZA

PRG, University of KwaZulu-Natal

N/A

- Co-Head, Pollution Research Group.

Dr. Kate Medicott

Geneva, CH

World Health Organization

N/A

- Team Leader, Sanitation.